

COGNITIVE ASPECTS



CHAPTER SIX

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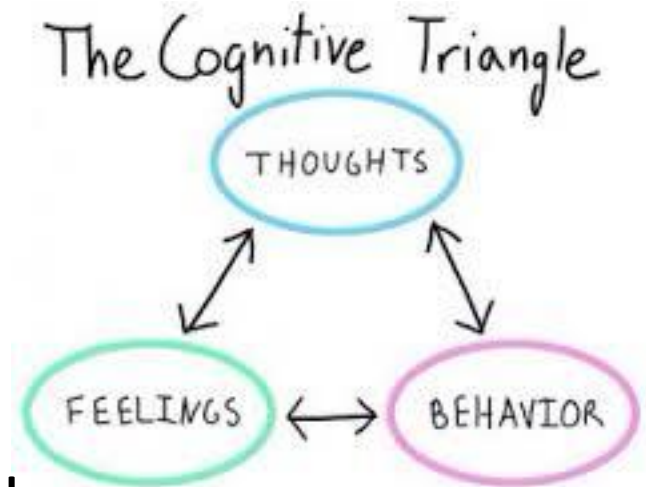
? cognitive

- Cognitive is how a person understands and acts in the world
- It is the set of abilities or processes that are part of nearly every human action
- Cognitive abilities are brain-based skills we need to carry out any task from the simplest to the most complex



COGNITIVE PROCESS

- Cognitive processes are processes that involve knowledge and how people use their knowledge.
- It involves thoughts, feelings and behaviours
- They include matters such as attention, memory, producing and understanding language, solving problems and making decisions.
- All these are very important for human behaviour.



COGNITIVE PROCESS

- Responsible for decisions – comparison and process of stimuli and selection of a response



COGNITIVE PROCESS

- There are 3 types of cognitive process :
 1. Mechanical – based on habits and repetition (eg. Walk, point, speak)



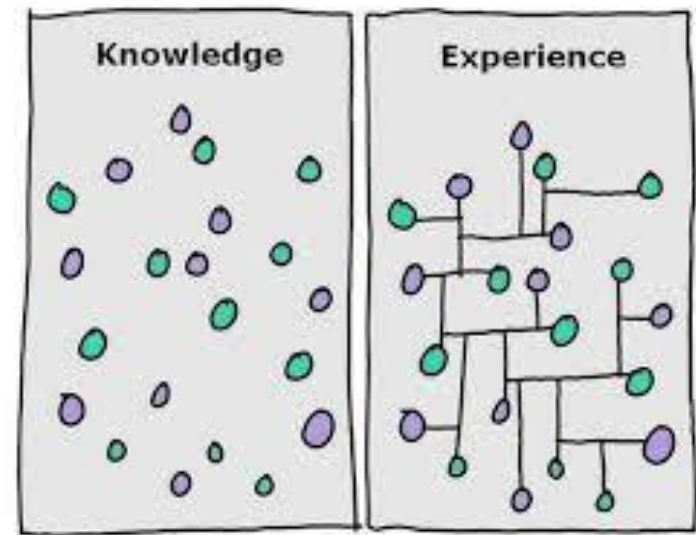
COGNITIVE PROCESS

- There are 3 types of cognitive process :
 1. Based on perception (eg. If there is an obstacle walk around it)
 2. Based on rules (eg. If there is an obstacle walk around it)
 3. Based on goals (eg. If there is an obstacle walk around it)



COGNITIVE PROCESS

- There are 3 types of cognitive process :
 1. Based on perception (problem solving)
 2. Based on logic (problem solving)
 3. Based on knowledge and experience (problem solving)



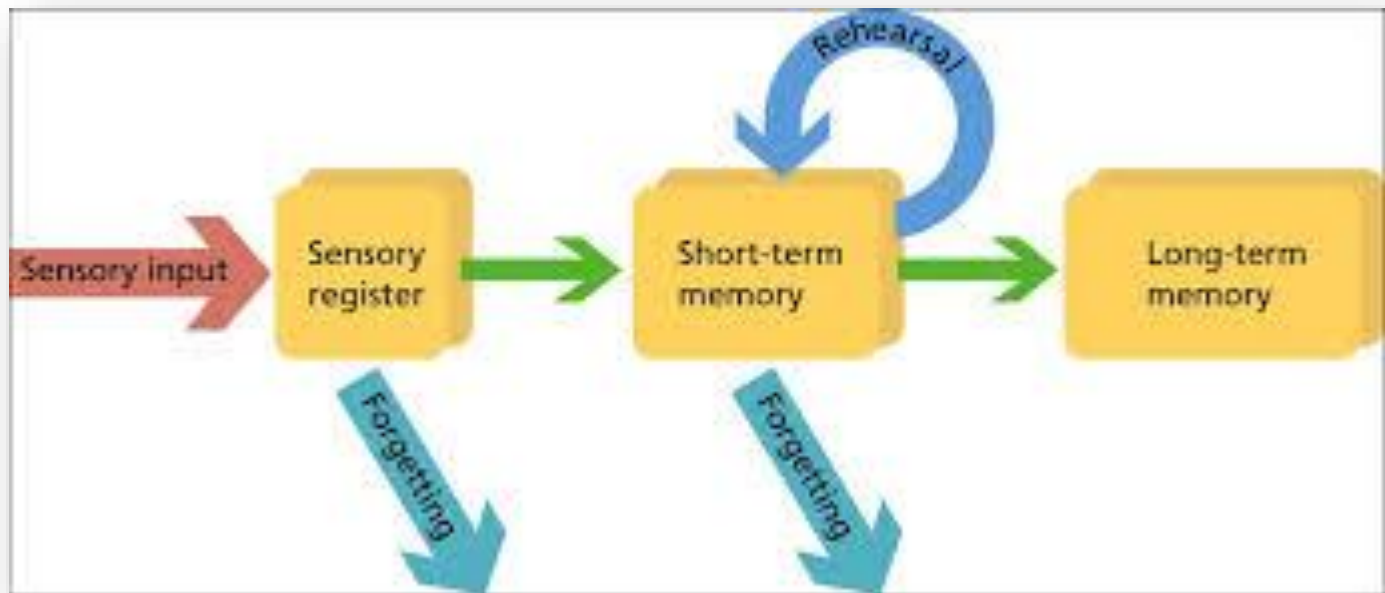
Cognitive process

- Memory and Learning



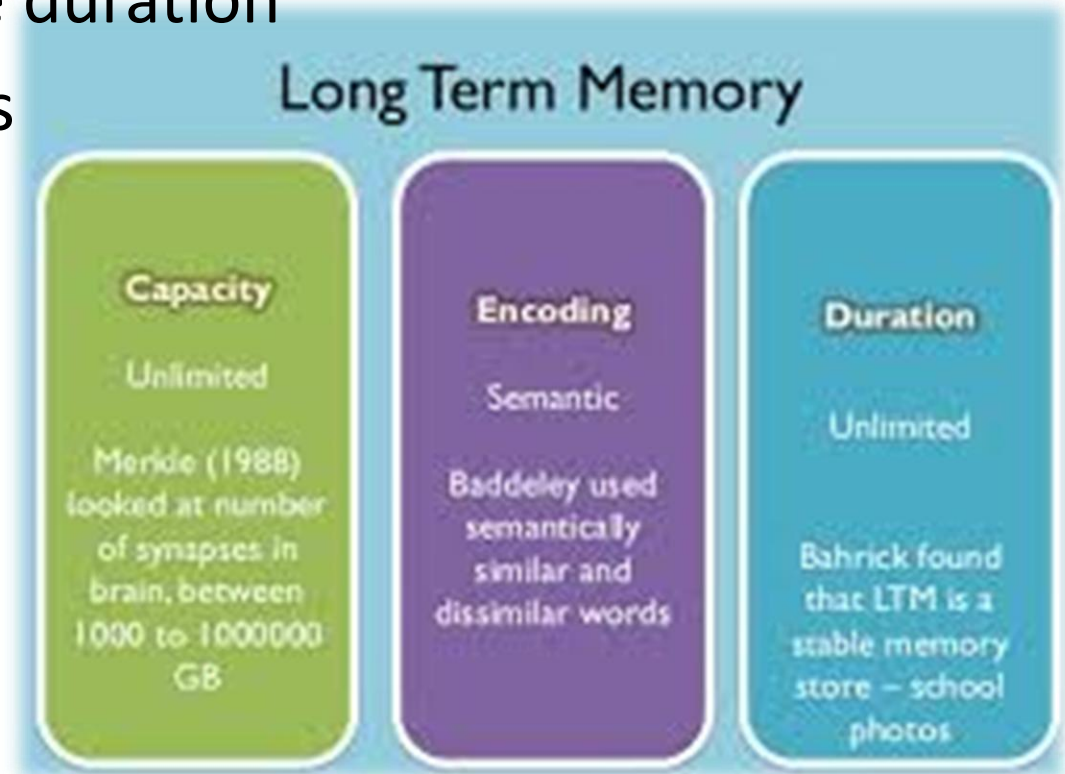
Cognitive process

- Short term memory
 - Working memory
 - Small capacity : 7 ± 2 items
 - Small storage duration(10-30s)



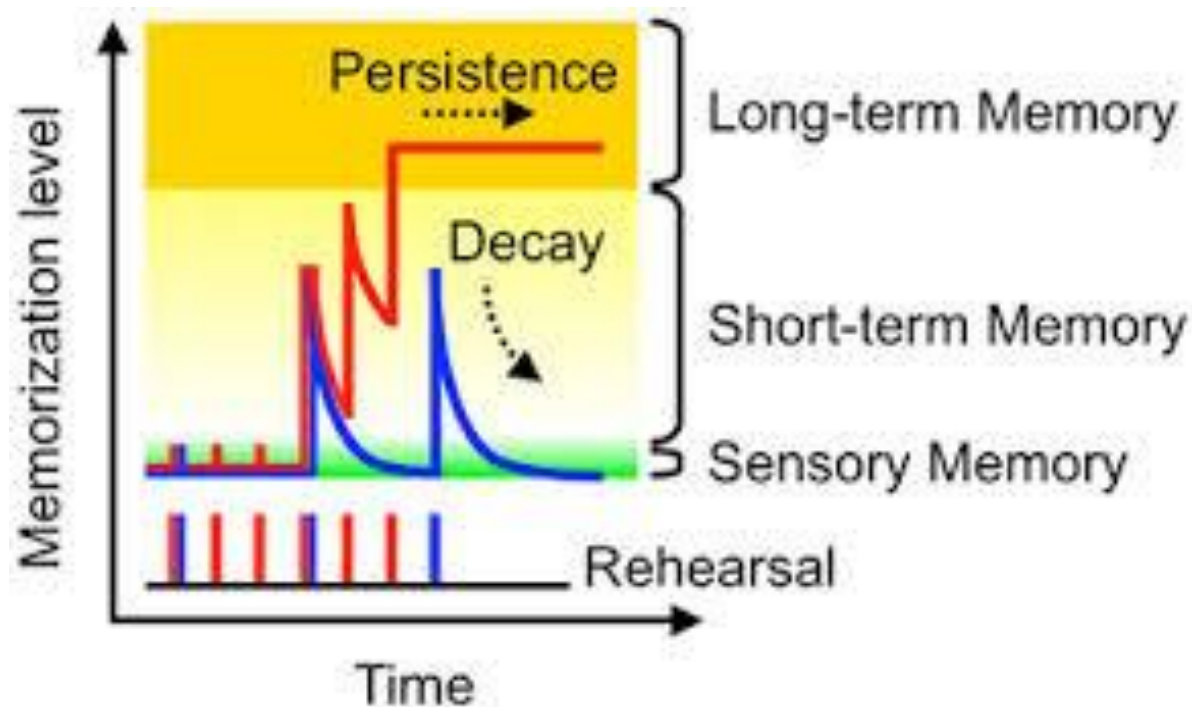
Cognitive process

- Long term memory
 - Infinite capacity
 - Unlimited storage duration
 - Associative access



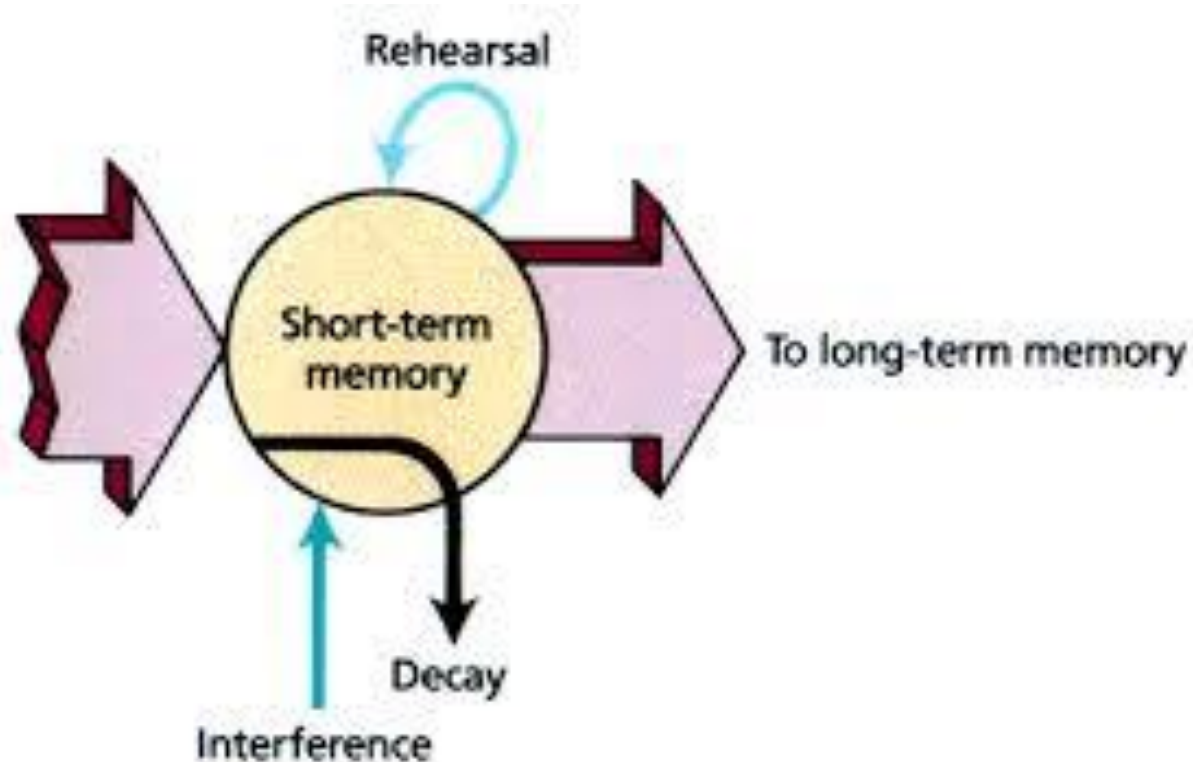
Cognitive process

- Learning and memorization by repetition
(short term -> long term)



Cognitive process

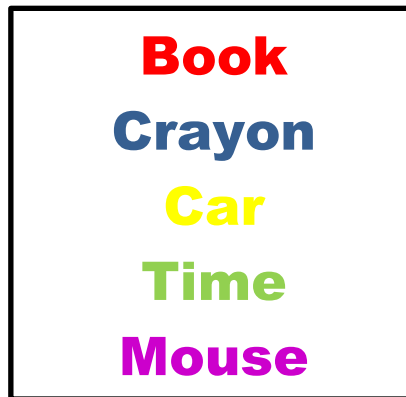
- Interference degrade short term memory faster



Cognitive process

Example of interference

- Test 1
- Identify the colour of the following words in order, as fast as possible



Cognitive process

- Interference between the main task (identify the colour) and cognitive process (read a word) – affects reaction time and error rate



Recognition over Recall

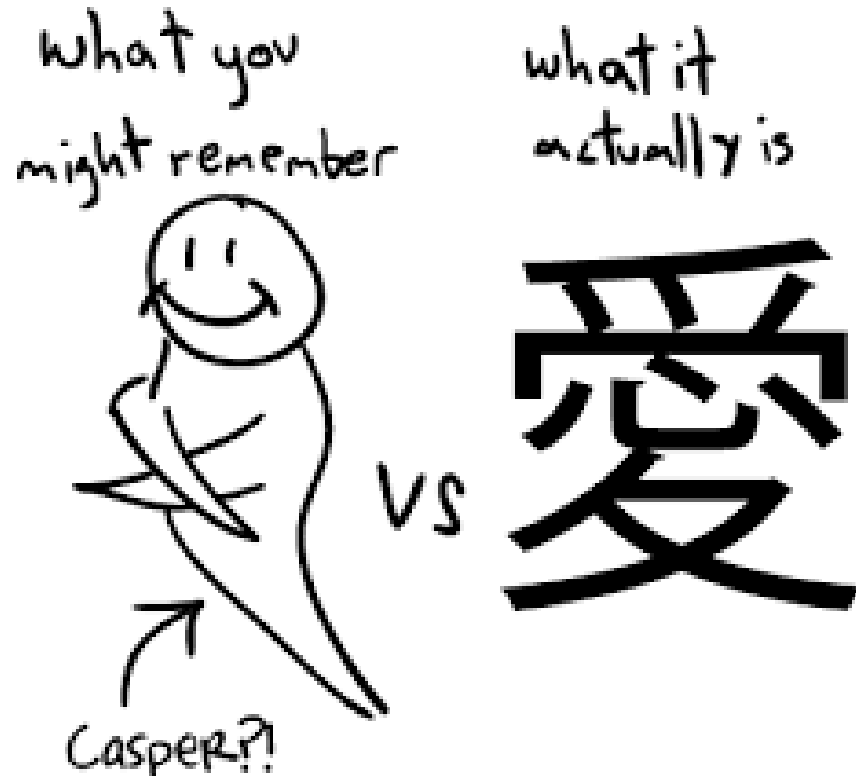
— ATTENTION / PERSUASION —

*It's easier to recognize things we have
previously experienced than it is to recall
those things from memory.*



Recall vs. recognition

- Recognition and recall are two ways that your brain is able to retrieve information that has been stored in your memory.



Recall vs. recognition

- Recognition needs a cue before it can fire, while recall does not require any specific cues to start the impulses.



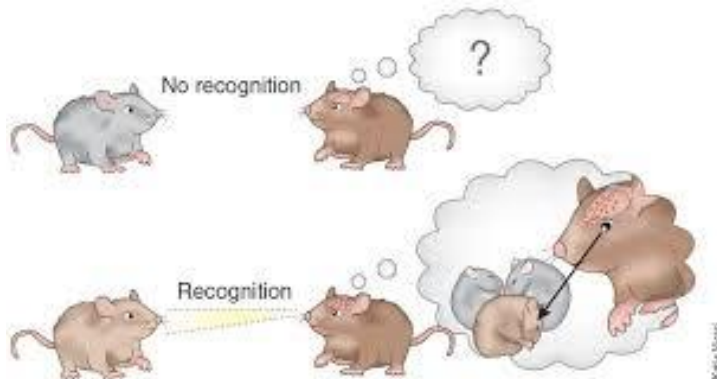
Recall vs. recognition

- Recognition is considered easier than recollection, since you have a trigger.



Recall vs. recognition

- Recognition is how you respond to a sensory cue. You look at something and your mind looks to see if what you are seeing in front of you matches anything that has been stored. If you notice a match, you are recognizing the information.

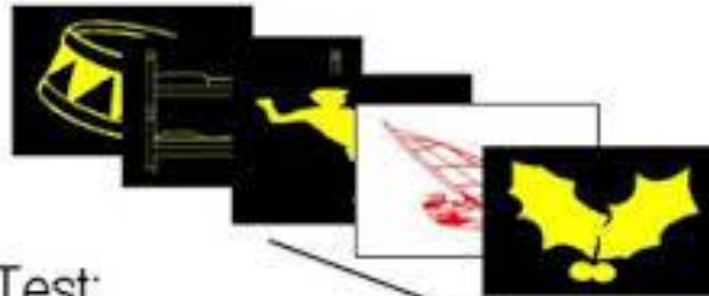


		
McDonalds Everyone knows the golden arches as McDonalds.	Nike We associate a tick icon with the brand Nike.	Apple We all know this is the icon on our iPhone and iPads.

Recall vs. recognition

- Recall is a way that you obtain information from your memory without having a cue to prompt the response. You have to essentially draft the information without any assistance.

Study:



Test:



Recall vs. recognition

- An example of recognition vs. recall in reality is taking a test. If the test is multiple choice, you will use recognition to find the answer. You look at one of the choices and your brain may recognize that one of the answers has already been stored. Alternatively, recall is used in the form of a short answer, fill in the blank or essay format.



Recall vs. recognition

- A police line-up is a classic exercise in recognition. You look at several people, and compare each to the person you saw commit the crime.



Recall vs. recognition

- Say that instead of looking at a line-up, you have to describe the person you saw to a sketch artist. This is an exercise in recall. The artist may try to help your recollection by asking questions, but ultimately you have to find the information yourself.



Recall vs. recognition

- When comparing recall vs. recognition, recall tends to be harder than recognition because there are no "hints" in place to aid you in finding the answer.



Recall vs. recognition

- Recall or retrieval of memory refers to the subsequent re-accessing of events or information from the past, which have been previously encoded and stored in the brain. Also known as remembering.

Memory Recall



Recall vs. recognition

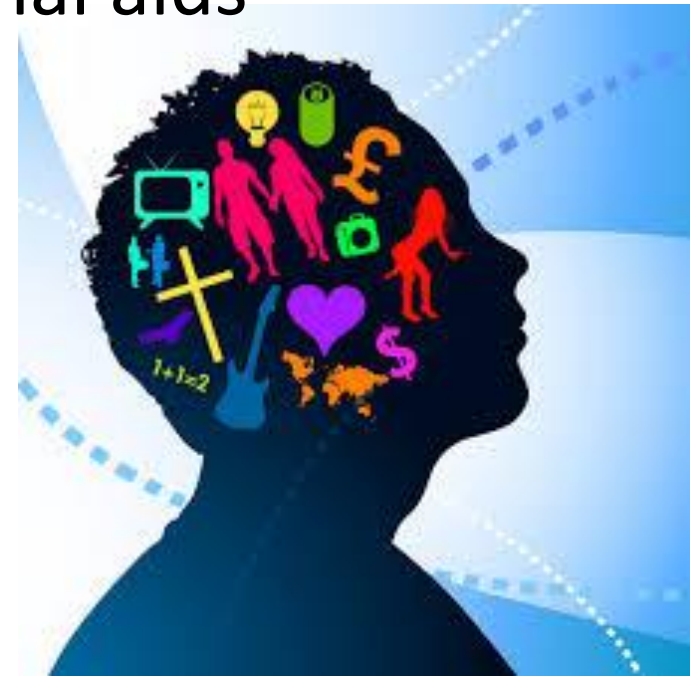
- During recall, the brain "replays" a pattern of neural activity that was originally generated in response to a particular event, echoing the brain's perception of the real event. In fact, there is no real solid distinction between the act of remembering and the act of thinking.



More on COGNITIVE

Externalization of cognition

- The use of the external world to achieve cognition
- Ways that people augment their normal cognitive processes with external aids
- Example:
 - External writings
 - Visualizations
 - Work spaces



Externalization of cognition

- Why?
 - To remind us :
 - that we need to do something,
 - what to do and
 - when to do them



Externalization of cognition

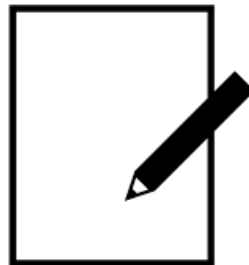
- Externalizing will reduce cognitive load
- Strategies that been developed to help remember difficult things
 - Example of difficult things to remember are:

- Birthdays
- Appointments
- Address



Externalization of cognition

- Computational offloading
 - Occurs when use a tool device in conjunction with an external representation to help us carry out a computation:
- Eg: paper and pen, calculator, spread sheet to solve a math problem



Externalization of cognition

- Try to calculate $234 \times 456 = ?$

Externalization of cognition

- Try to calculate $234 \times 456 = ?$
 - In your mind



Externalization of cognition

- Try to calculate $234 \times 456 = ?$

– On paper



Externalization of cognition

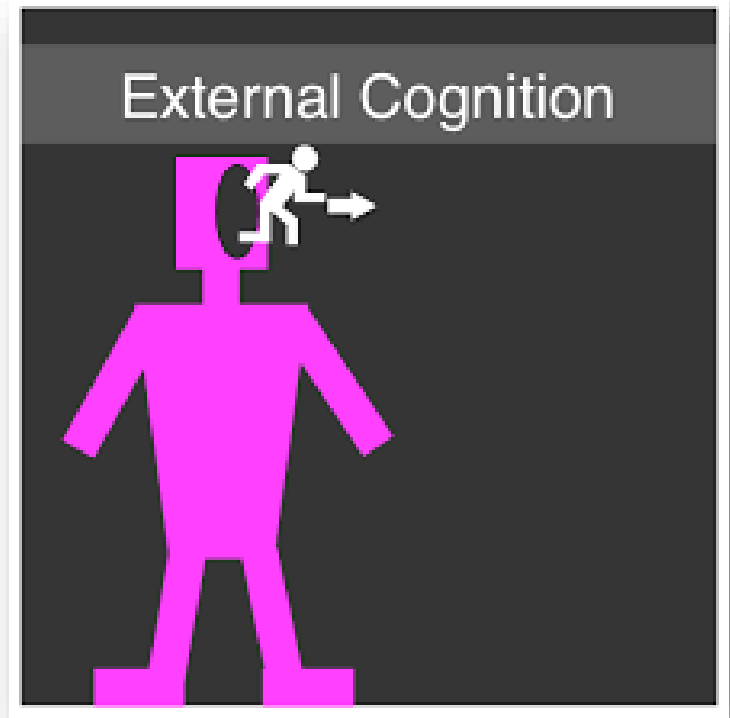
- Try to calculate $234 \times 456 = ?$

— With calculator



Externalization of cognition

Another way in which we externalize our cognitions is by modifying representations to reflect changes that are taking place that we wish to mark.



Externalization of cognition

- Two kinds of modifications
 - 1. Annotating** - modifying existing representations through making marks to mark progression in tasks
 - Eg: crossing off, ticking, underlining



Externalization of cognition

- Two kinds of modifications
 2. **Cognitive tracing** - manipulating items into different orders or structures that are easier to remember
 - Eg: playing scrabble, playing cards, history



Menus stimulate *cognitive*

*Good*Menus

vs

*Bad*Menus

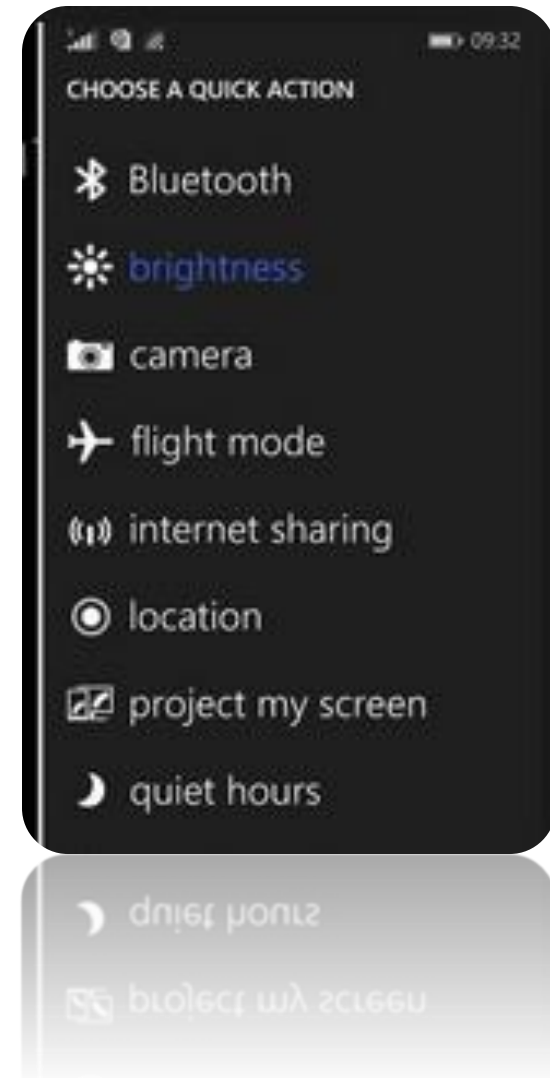
Menus



- How long does it take to find an item in a menu from the figure above?

Menus

- If the items are ordered (eg. Alphabetically), the choice time is approximated by Hick's Law (logarithmic) -> expert use



Menus

- If the items are randomly ordered and the user does not know their position, they need to search for the target in a linear way (rather than logarithmic) -> novice use

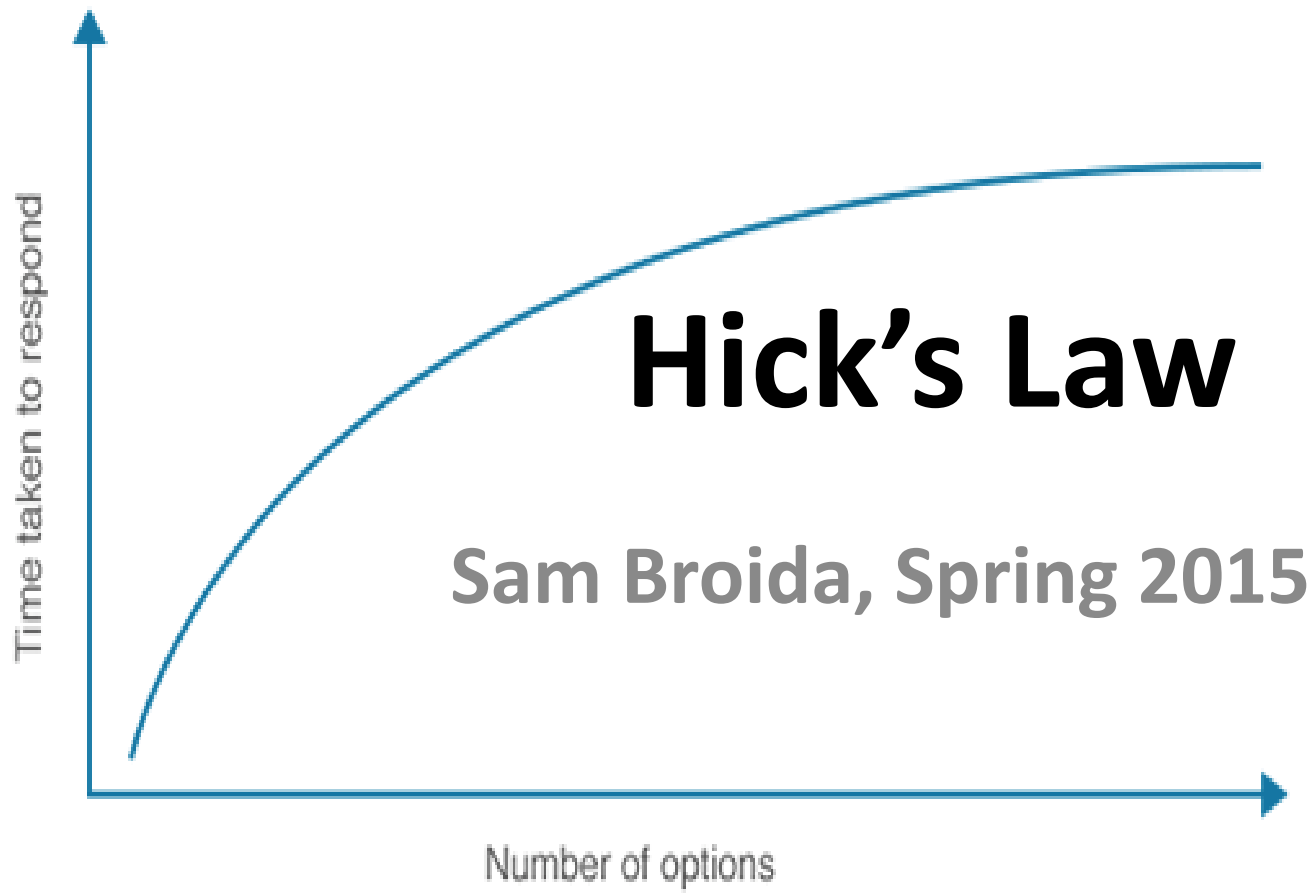


???

Designing Menus

Guidance in
designing menus

1.



2.

Fitts' Law



Previous chapter

Covered in

3.

Steering law

1.

Hick's Law

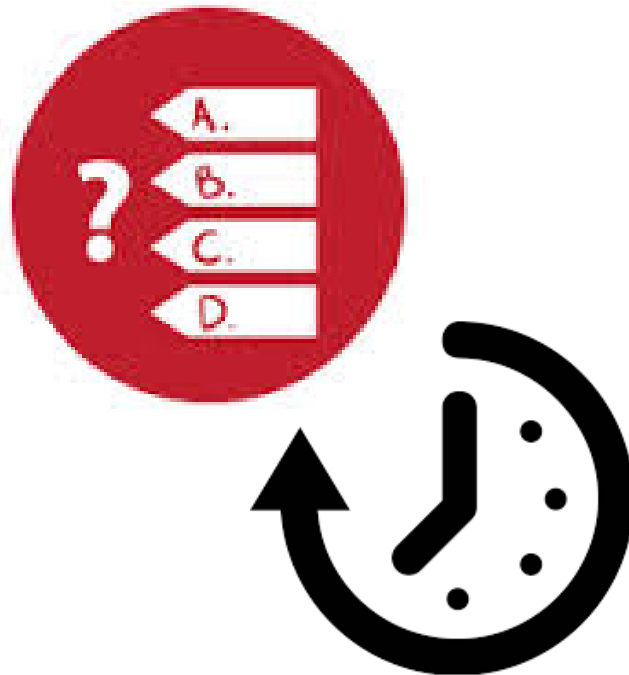
Hick-Hyman Law

- William Edmund Hick
 - British Psychologist
 - 1952
- Ability of a human to quickly predict an outcome based on the number of simultaneous stimuli



Hick-Hyman Law

- This law describes the time it takes for a person to make a decision as a result of the possible choices he or she has



Practical Example

- “A classic example of Hick’s Law in action is a study testing the effects of jam options in a busy supermarket. The study used two different jam tasting displays. One with 24 different jams, and the other with only six.



Practical Example

- Results showed that with the 24 option display, 60% of people passing tried the jam with 3% of them purchasing. With the 6 jam display, a lesser 40% of people stopped but **almost 30% purchased**





Hick-Hyman Law

- increasing the number of choices will increase the decision time logarithmically.



- The Hick–Hyman law assesses cognitive information capacity in choice reaction experiments.



Hick-Hyman Law

- The amount of time taken to process a certain amount of bits in the Hick–Hyman law is known as the **rate of gain of information**.



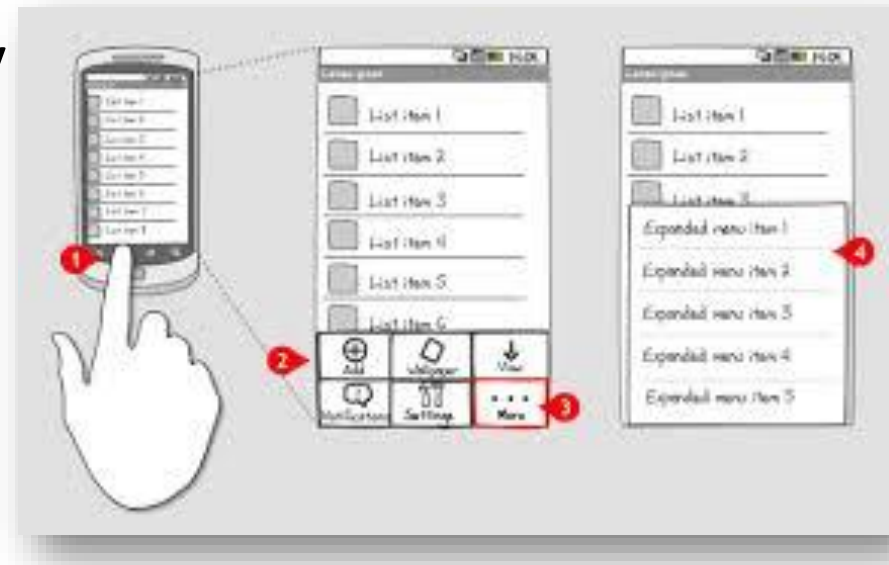
Hick-Hyman Law

- Hick's law is sometimes cited to justify menu design decisions.



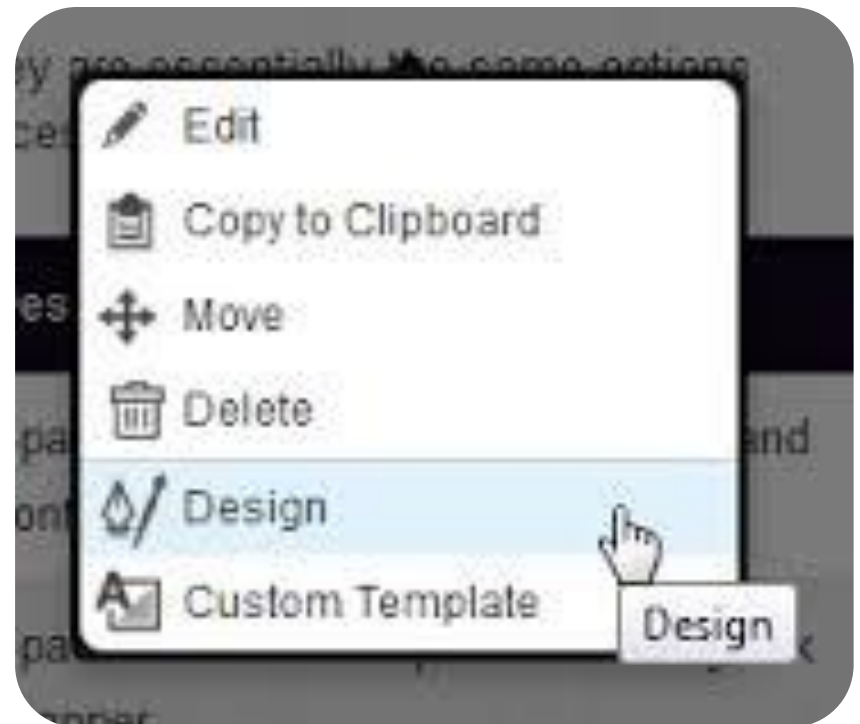
Hick-Hyman Law

- For example,
 - to find a given word (e.g. the name of a command) in a randomly ordered word list (e.g. a menu),
 - scanning of each word in the list is required, consuming linear time, so Hick's law does not apply.



Hick-Hyman Law

- However, if the list is alphabetical and the user knows the name of the command, he or she may be able to use a subdividing strategy that works in logarithmic time.



Hick-Hyman Law

....1,2,3,4 steps

- Menu Design task consist of four basic steps:
 1. identify a problem or goal,
 2. assess the available options to solve the problem or achieve the goal,
 3. decide on the options,
 4. implement the options.

Hick-Hyman Law

Hick's Law applies to the third step:
(decide on an option).



Hick-Hyman Law

However, the law does not apply to decisions that involve significant levels of searching, reading, or complex problem solving.

- For example, a complex task requiring reading sentences and intense concentration with three options can easily take longer than a simple stimulus-response task with six options.



Hick-Hyman Law

- Therefore, Hick's Law is most applicable for simple decision-making tasks in which there is a unique response to each stimulus.
 - For example, if A happens, then push button 1, If B happens, then push button 2. The law is decreasingly applicable as the complexity of tasks increases



Hick-Hyman Law

- Designers can improve the efficiency of a design by understanding the implications of Hick's Law.



Hick-Hyman Law

- For example, the law applies to the

- design of software menus,



- control displays,



- way finding layout



Hick-Hyman Law

- For example, the law applies to the

— signage,



— emergency response training—



as long as the decisions involved are simple.

Hick-Hyman Law

- As the complexity of the tasks increases, the applicability of Hick's Law decreases.
- For example, Hick's Law does not apply to
 - complex menus or hierarchies of options.

FOR YOUR HANDS	
Choose between -	
High Fiber Brown Loaf or	
Easy Master Blend Masala Loaf	
Paneer Tikka Sandwich	135
Kofta Chilly Mayo Sandwich	135
Cheese & Herb Sandwich	135
BBQ Chicken Sandwich	145
Masala Roast Chicken Sandwich	145
Smoked Chicken Sandwich	145
WITH A KNIFE & FORK	
Butter Croissant	50
Chocolate Croissant	75
Spicy Croissant	65
Paneer Tikka Croissant	135
Kofta Chilly Mayo Croissant	135
Cheese & Herb Croissant	135
BBQ Chicken Croissant	145
Masala Roast Chicken Croissant	145
Smoked Chicken Croissant	145
Vegetable Puff	65
Chicken Jungli Puff	75
Vegetarian Quiche	65
Chicken Quiche	75
SOMETHING LIGHT	
Toast with Butter / Jam	55
Bowl of Chips -	
Masala, Tomato or Plain	50
Chocolate Chip Cookie	45
Blueberry Muffin	75
Chocolate Chip Muffin	75
SOMETHING HOT	
Vegetarian Lasagna	160
Chicken Lasagna	170
Fusilli Al Funghi Pasta	160
Chicken Penne Pasta	170
CHOCOLATE TREATS	
Chocolate Marble Brownie	120
Blueberry Chocolate Mousse	110
Sydney Chocolate Slice	125
Cashew & Hazelnut Verrine	125
SUGAR HIT	
Date & Walnut Cake	70
Carrot Cake	65
Fruit Cheesecake	125
Tiramisu	125
Red Velvet Cake	125
SIGNATURE COFFEES	
Cappuccino	100 120
Koalaccino	120 140
Flat White	100 120
Americano/Long black	100 120
Miniccino	90 110
Affogato	110 130
Café Latte	110 130
Café Mocha	110 130
Chai Latte	100 120
Espresso/Short black	80 100
Macchiato	80 100
Piccolo Latte	80 100
Good Co. Flavours	30
Extra Shot	40

Hick-Hyman Law

- Menu selection of this type is not a simple decision-making task since it typically involves:

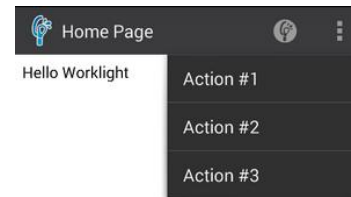
- reading sentences,



- searching and



- scanning for options, and

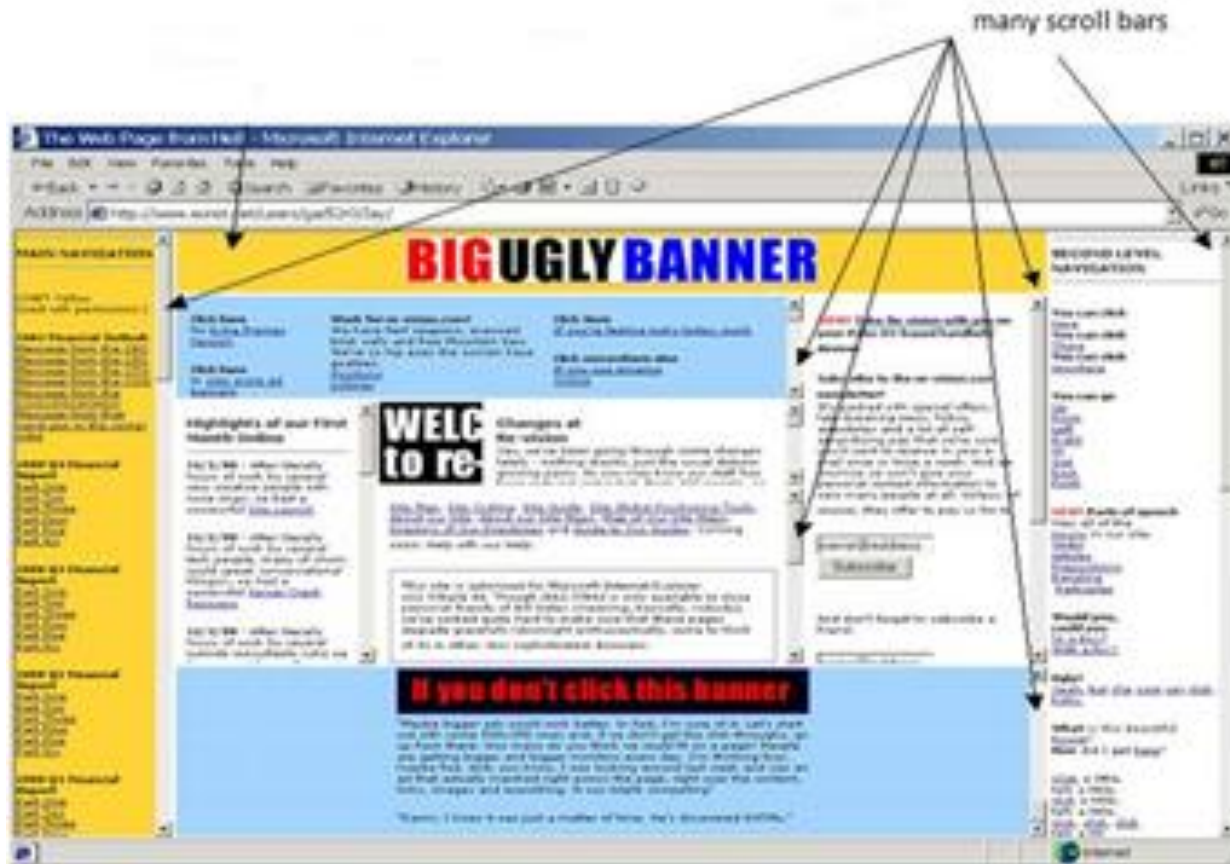


- some level of problem solving.



Hick-Hyman Law

- Consider Hick's Law when designing systems that involve decisions based on a set of options.



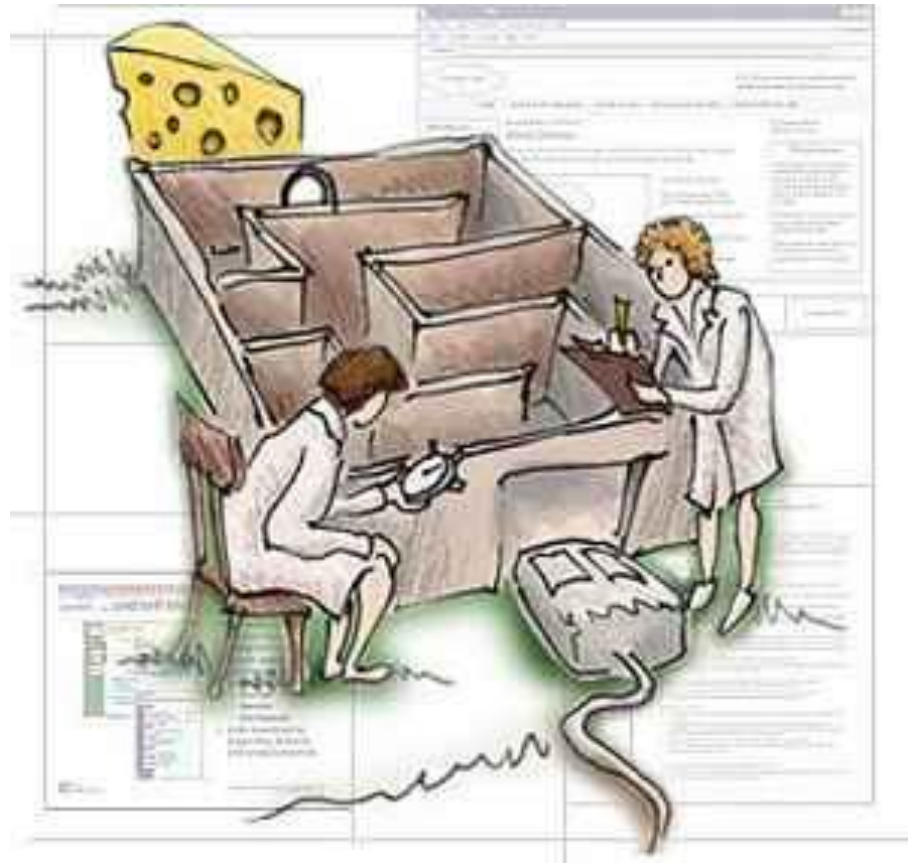
Hick-Hyman Law

- When designing for time-critical tasks, minimize the number of options involved in a decision to reduce response times and minimize errors.



Hick-Hyman Law

- When designs require complex interactions, do not rely on Hick's Law to make design decisions; rather, test designs on the target population using realistic scenarios.



Hick-Hyman Law

- In training people to perform time-critical procedures, train the fewest possible responses for a given scenario.
- This will minimize response times, error rates, and training costs.



Hick-Hyman Law

- Useful tips based on Hicks Law for web-page design
 1. Distinguish links using colors



Hick-Hyman Law

- Useful tips based on Hicks Law for web-page design
 2. Use consistent and familiar layouts to reduce the reaction time



Hick-Hyman Law

- Useful tips based on Hicks Law for web-page design
 3. Use techniques (e.g. lines, colors) to distinguish related blocks of information (that helps in selecting alternatives quickly) to reduce overall reaction time



3.

Steering law

Steering law (movement vs. menus)

- The steering law in human–computer interaction and ergonomics is a predictive model of human movement that describes the time required to navigate, or steer, through a 2-dimensional tunnel.



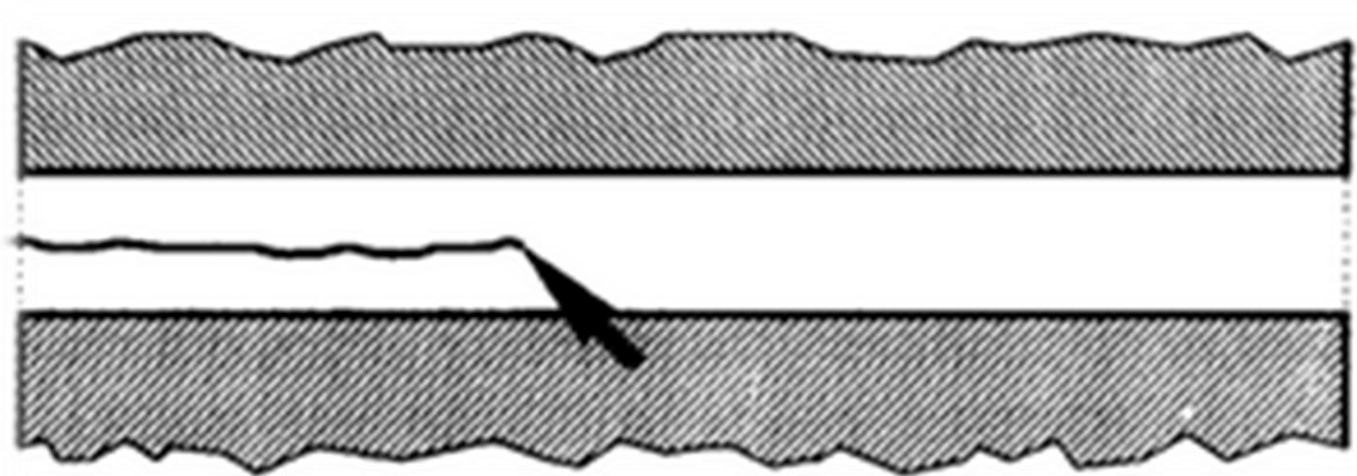
Steering law (movement vs. menus)

- The tunnel can be thought of as a path or trajectory on a plane that has an associated thickness or width, where the width can vary along the tunnel.



Steering law (movement vs. menus)

- The goal of a steering task is to navigate from one end of the tunnel to the other as quickly as possible, without touching the boundaries of the tunnel.



Steering law (movement vs. menus)

- A real world example that approximates this task is driving a car down a road that may have twists and turns, where the car must navigate the road as quickly as possible without touching the sides of the road.



Steering law (movement vs. menus)

- The steering law predicts both the instantaneous speed at which we may navigate the tunnel, and the total time required to navigate the entire tunnel.



Steering law (movement vs. menus)

- In this context, the steering law is a predictive model of human movement, concerning the speed and total time with which a user may steer a pointing device (such as a mouse or stylus) through a 2D tunnel presented on a screen (i.e. with a bird's eye view of the tunnel), where the user must travel from one end of the path to the other as quickly as possible, while staying within the confines of the path.



Steering law (movement vs. menus)

- One potential practical application of this law is in modeling a user's performance in navigating a hierarchical cascading menu.



Steering law (movement vs. menus)

- Many researchers in human–computer interaction, including Accot himself, find it surprising or even amazing that the steering law model predicts performance as well as it does, given the almost purely mathematical way in which it was derived. Some consider this a testament to the robustness of Fitts's law.



Steering law (movement vs. menus)

- In its general form, the steering law can be expressed as

$$T = a + b \int_C \frac{ds}{W(s)}$$

- where T is the average time to navigate through the path, C is the path parameterized by s , $W(s)$ is the width of the path at s , and a and b are experimentally fitted constants. In general, the path may have a complicated curvilinear shape (such as a spiral) with variable thickness $W(s)$.



Steering law (movement vs. menus)

- Simpler paths allow for mathematical simplifications of the general form of the law. For example, if the path is a straight tunnel of constant width W , the equation reduces to

$$T = a + b \frac{A}{W}$$

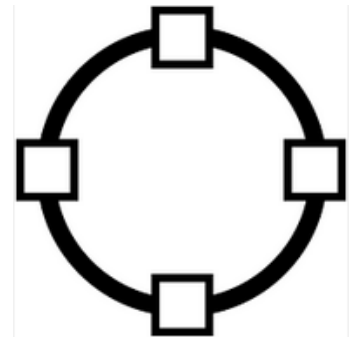


- where A is the length of the path. We see, especially in this simplified form, a *speed-accuracy* tradeoff, somewhat similar to that in Fitts's law.

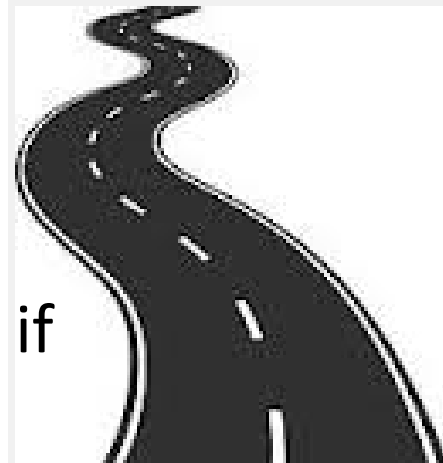
Steering law (movement vs. menus)

- We can also differentiate both sides of the integral equation with respect to s to obtain the local, or instantaneous, form of the law:

$$\boxed{\frac{ds}{dT} = \frac{W(s)}{b}}$$

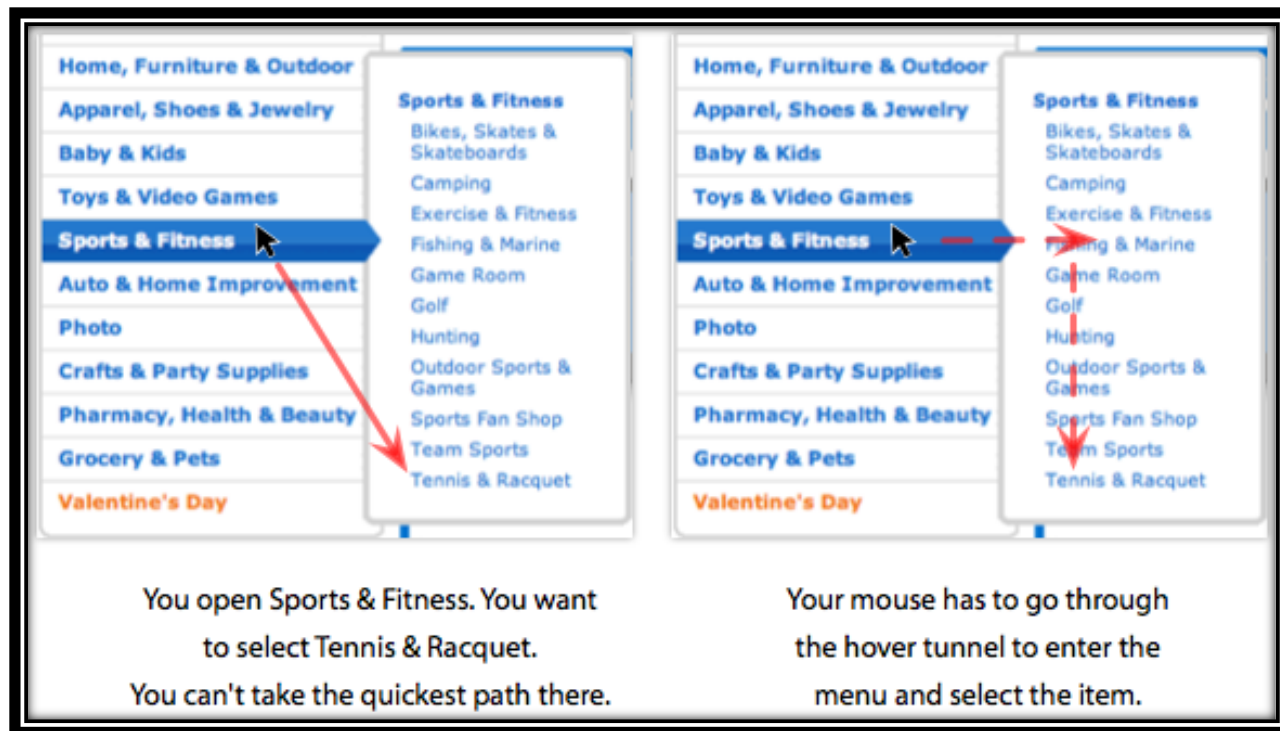


- which says that the instantaneous speed of the user is proportional to the width of the tunnel. This makes intuitive sense if we consider the analogous task of driving a car down a road: the wider the road, the faster we can drive and still stay on the road, even if there are curves in the road.



Steering law (movement vs. menus)

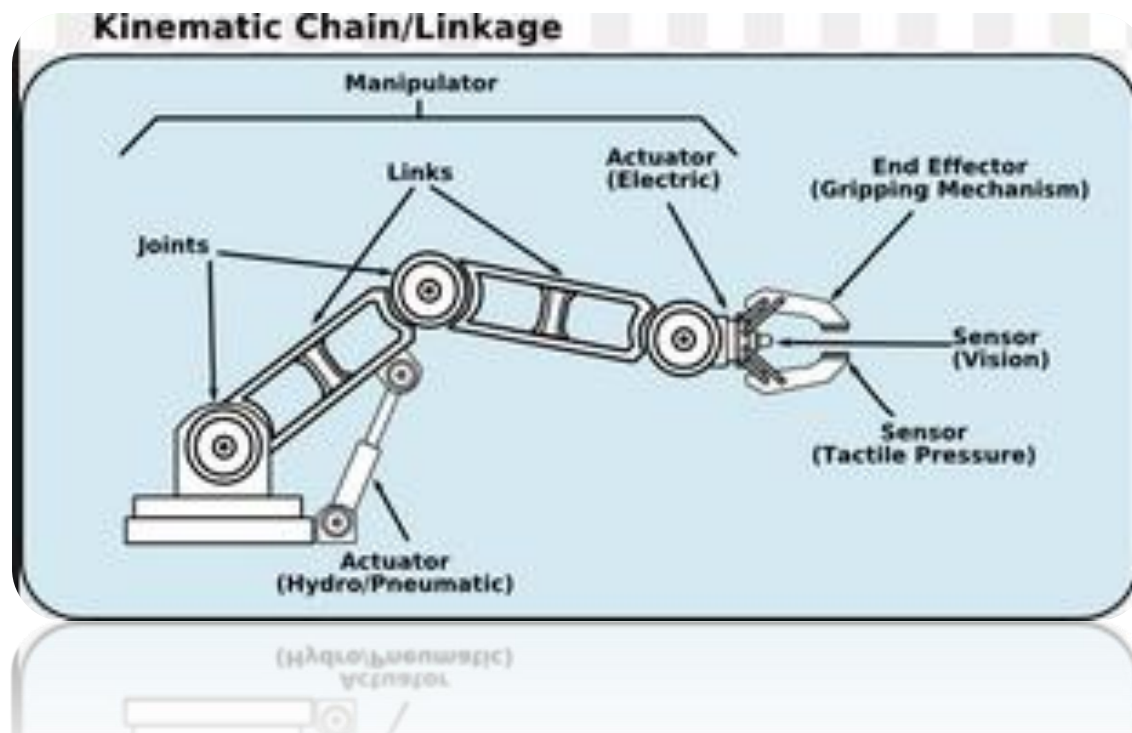
- Example of Steering Law



Other consideration in designing
menus

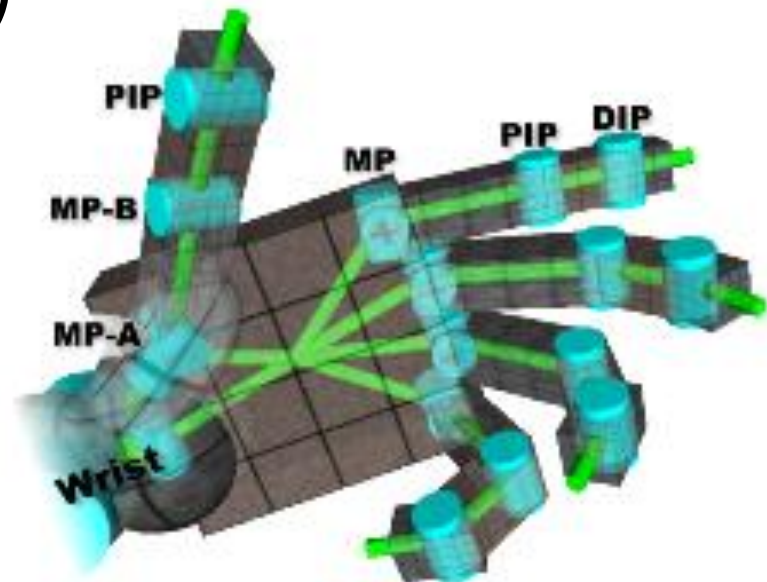
Kinematic Chain Model (Guiard, 1987)

- The kinematic chain is a sequence of abstract motors
- Eg: shoulder -> elbow -> wrist -> finger



Kinematic Chain Model (Guiard, 1987)

- For each link in the sequence (eg. Wrist -> finger) the 2nd element (finger) defines its movement with respect to the movement of the 1st element (wrist)



Kinematic Chain Model (Guiard, 1987)

- Hypothesis: bi manual movement acts as a kinematic chain
 - Non-dominant hand -> dominant hand



Kinematic Chain Model (Guiard, 1987)

– Principles :

- The dominant hand moves in the references frame established by the non-dominant hand
- Asymmetry in the chain of actions. Movements of the dominant hand more precise
- The non-dominant hand precedes the dominant hand. Eg. Left hand positions a paper and the right hand starts writing



Kinematic Chain Model (Guiard, 1987)

- Laterality of motor control
 - Classical psychology:
 - “the left hand is a bad right hand”
 - Observations of bimanual control:
 - the two hands have different roles



Kinematic Chain Model (Guiard, 1987)

- Kinematic chain:
 - Non-dominant hand: distal control
 - • Acts first
 - • Establishes the frame of reference (context) for the dominant hand
 - • Movements do not need to be precise



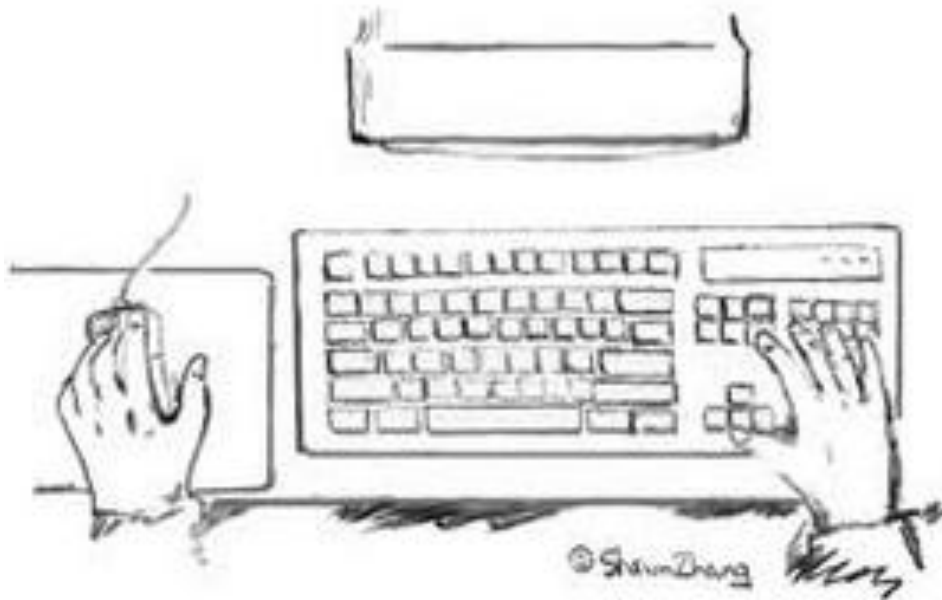
Kinematic Chain Model (Guiard, 1987)

- Kinematic chain:
 - Dominant hand: proximal control
 - • Acts after the non-dominant hand, within the frame of reference it establishes
 - • Precise movements



Kinematic Chain Model (Guiard, 1987)

- Falsification :
 - Some tasks are more efficient when the hands have symmetric roles



Good example of menus design

Menus in MAC OS X





End of Lecture